



Muhammad Jumi'at Mokhtar

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ABOUT ME

I am a IoT Solution Engineer with hands-on experience in designing and delivering end-to-end technical solutions that bridge real-world operations with digital systems. My background spans system integration, backend services, API-based communication, and data-driven platforms, with strong foundations in JavaScript, REST APIs, and SQL.

I have worked closely with cross-functional teams and stakeholders to translate business and operational needs into reliable, scalable solutions. My experience includes estimating implementation effort, designing system architecture, integrating multiple data sources, and supporting deployment and testing in production environments.

With a strong engineering mindset and exposure to IoT and real-time systems, I bring a practical approach to problem-solving—focused on reliability, clarity, and impact. I am highly adaptable, enjoy learning new technologies, and thrive in roles that require both technical depth and clear communication with customers and non-technical teams.

EDUCATION & CERTIFICATION

Universitas Lambung Mangkurat | *Bachelor of Physics*

Banjarbaru, Indonesia | **26/08/2017 – 30/09/2022**

- Specialization in Electronics and Instrumentation
- Active member and committee of student organizations, including the Physics Student Association, Islamic Spiritual Organization, and Student Representative Council
- 1st Place, University-Level Physics Competition (2021) and Bronze Medalist, National-Level Physics Competition (2020)
- Teaching Assistant for laboratory courses: Basic Physics, Sensor Systems, Measurement Systems, Digital Electronics, Industrial Instrumentation, and Microprocessor & Interfacing
- **GPA:** 3.62/4.00

IoT Network Engineer | *PT. LSP Telekomunikasi Digital Indonesia*

Jakarta, Indonesia | **13/05/2024 – 17/05/2024**

- Training provided by Floatway Systems, covering integration, testing, operation, OTA firmware installation, and security for IoT devices.
- Certified BNSP on: May 30, 2024

SKILLS

- **Software :** *Microsoft Applications (Word, Excel, PowerPoint); Development Tools (Visual Studio Code, Arduino IDE, STM32CubeIDE, ESP-IDF, Thonny IDE); Operating Systems (Linux, Windows); Collaboration Tools (Jira, Notion)*
- **Programming & Integration :** *Backend & Web Technologies (JavaScript, TypeScript, Node.js, PHP, SQL, Python, Golang); API & Data Integration (MQTT, REST API, Webhooks); GUI (C & C++); Low-Code & Automation (Node-RED); Version Control (Git)*
- **Systems & Communication:** *Networking & Protocols (Modbus TCP/IP, UDP, FTP); Database Systems (MySQL, PostgreSQL)*
- **Embedded & IoT (Supporting Skills):** *Microcontrollers (Arduino MCU Family, ESP32, STM32, Raspberry Pi Pico); Mini PC Platforms (Raspberry Pi, LattePanda); Connectivity Technologies (Ethernet, WiFi, BLE, RS485, CAN-BUS, NB-IoT); PLC (Schneider)*

EXPERIENCE

Program Penelitian Dosen | *Research Member*

Banjarbaru, Indonesia | 21/03/2020 - 20/12/2022

- Grant Research on Oil Palm: Conducted under the BDPKS research grant. Developed and calibrated temperature measurement tools for monitoring oil palm heating inside dump trucks (2020).
- Designed and implemented embedded systems using ESP8266 and ESP32 microcontrollers for environmental monitoring in agricultural research. Integrated more than 5 sensor environmental sensors (light intensity, soil temperature, soil moisture, air temperature, and humidity) for data acquisition and calibration.
- Developed a web-based monitoring system using PHP and MySQL, featuring a real-time dashboard, data logging, and admin interface. Established RESTful API and MQTT communication protocols for real-time data monitoring with <2 s data latency transfer between 99% IoT device connectivity uptime and the web server.
- Delivered Arduino-based robotics workshops to 30+ students and authored a beginner-friendly Arduino learning module for MAN 4 Banjar School.

PT. Respati Solusi Teknologi (RESTEK) | *Engineer*

Jakarta, Indonesia | 06/04/2023 – 10/04/2025

- Developed backend software for monitoring and controlling the Automatic Train Wash Plant (ATWP) of LRT Jabodebek, integrating PLC systems via Modbus TCP/IP protocol. reducing manual oversight by 40%. Built server-side applications using JavaScript and PostgreSQL for real-time data acquisition 24/7 operation uptime, and process control.
- Designed and developed PCB controller modules and firmware for RGB LED Matrix Running Text displays, programmed using Arduino IDE, deployed in >50 train units in LRT Jabodebek, Commuter Line, and Feeder trains KCIC. Improving system stability by 30%.
- Designed an ESP32–Raspberry Pi–based monitoring system integrating GPS, accelerometer, and magnetometer sensors, using UDP and 4G connectivity for real-time trainset tracking enabling continuous remote with <2 s update intervals and 98% data reliability.

PT. Aman Media Interaktif (Amani Tech) | *IoT Presales Engineer*

Jakarta, Indonesia | 05/06/2025 – Present

- The validator ensures the consistency, accuracy, and integrity of critical driving behavior and driver condition data, helping to identify anomalies, validate event triggers, and improve overall system reliability.
- Led Amani Visual Analytic platform development using OpenCV and YOLO for real-time image and video validation, reducing manual verification time by 60% (processing photos in ~5s and videos in ~6min per file).
- Collaborated with AI and infrastructure teams to optimize data pipeline throughput, improving system reliability and event validation accuracy by 25%.
- Developed a desktop middleware system for Battery charging. using serial communication protocol to convert byte data from old apps to new device battery charging using protocol SCPI.
- Led installation, calibration, and commissioning of a locomotive fuel monitoring system using Teltonika FMC650, dual fuel level sensors, and CAN Bus display units, enabling real-time monitoring of fuel volume and temperature through integrated telematics infrastructure.

PORTFOLIO

Agricultural Monitoring Measurement System | *Undergraduate Thesis*

Banjarbaru, Indonesia | 09/03 - 14/07/2022

- Designed and calibrated an agricultural monitoring system measuring light intensity, soil moisture, air temperature, and humidity
- Developed a PC-based monitoring application with real-time data visualization and automated data logging to spreadsheet formats
- Implemented end-to-end data acquisition and monitoring workflows, supporting analysis and system validation

Embedded & IoT System Development | *Self-taught*

Jakarta & Banjarbaru, Indonesia | 21/05/2022 - Present

- Developed real-time environmental monitoring integrating 10+ sensors tools for various applications : peatlands (oxygen levels, temperature, humidity) using Arduino and SIMCom 800L (GPRS module); fish ponds (water turbidity, temperature, humidity) using ESP8266; and home conditions (temperature, humidity) using RPi Pico, all with real-time data transmission capabilities.

- Designed and developed industrial IoT devices based on STM32 and ESP32 platforms, including Ethernet communication (W5500, ENC28J60), LWIP TCP/IP networking with <1% packet loss, MQTT telemetry, FreeRTOS-based firmware, and cellular connectivity using Quectel EC600K 4G LTE modems.
- Designed and implemented industrial HMI solutions for real-time locomotive fuel monitoring, integrating Eurosens Dominator fuel level sensors and flow meters over CAN Bus J1939 with ESP32-S3 7 inch displays using LVGL.
- Developed embedded display drivers and network utility devices, including STM32 graphical LCD drivers, ESP-IDF-based ePaper information displays, and portable diagnostic devices supporting Ethernet, WiFi AP/STA, and 4G LTE connectivity through browser-based interfaces.
- Developed a measurement tool using the RTOS library on ESP32, enabling it to function as both a web server and an MQTT client for sending data to a database and Node-RED, with a prototype server implemented on RPi.
- Developed a Python-based parser converting .BIN files from the Train Information System (TIS) into CSV datasets containing physical parameters, GPS coordinates, and train conditions, enabling real-time cloud monitoring and reducing manual data analysis time by 75%.
- Integrated Schneider M221 PLC with ultrasonic sensors and CCTV to enable remote monitoring of the Current Collector Device (CCD) Train, reducing on-site inspection frequency by 60% and improving maintenance efficiency.
- Developed a real-time weight-sensor data processing system with FTP image backup and camera integration, enhancing data traceability and reducing manual reporting time by 40%.

Web System Development | Self-taught

Jakarta, Indonesia | 01/01/2025 - Present

- **Dashboard Train Moving Monitoring System:** Developed a real-time monitoring dashboard for train movement and condition tracking. Implemented RESTful APIs, database integration, and real-time data updates with low-latency synchronization between edge systems and cloud services. (<https://trainmon.rsteknologi.com/train-monitoring>) (PHP, REST API, MySQL)
- **J2ME Retro Archive Platform:** Built a web-based archival platform for classic J2ME mobile games, featuring structured content management, fast client-side rendering, and scalable backend services for handling large digital collections. (<https://j2me-retro.mjmokhtar.cloud>) (Vue.js, Bun, TypeScript, PostgreSQL)
- **Company Profile & Project Showcase Website:** Developed a company profile website to present services and project portfolios, focusing on clean UI, modular backend structure, and maintainable deployment for long-term use. (<https://toaster.mjmokhtar.cloud>) (Node.js, MySQL)